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GR-CLASS Backend — Complete API Reference

For: Frontend & QA/Tester Teams

Base URL: [/api/v1](#)

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1. Roles Overview

Role	Code	Description
System Admin	ADMIN	Full system control — user management, configuration, audit logs.
General Manager	GM	Business approvals, surveyor assignment, certificate issuance.
Team Manager	TM	Operations oversight, survey finalization, certificate lifecycle.
Technical Officer	TO	Document verification, technical review post-survey.
Surveyor	SURVEYOR	Field inspection — checklist, GPS, photo, evidence capture.
Client (Vessel Owner)	CLIENT	Service requester — creates jobs, views certificates, pays invoices.

2. Auth Module

Why: Handles login, logout, password management, and token refresh for all users.

Base Path: [/api/v1/auth](#)

#	Method	Endpoint	Access	Description
1	POST	/login	 Public	Authenticate with email + password. Returns access + refresh tokens. Rate-limited (10 attempts / 15 min).
2	POST	/logout	 Any Authenticated	Invalidates current access token (adds to blacklist).
3	POST	/refresh-token	 Public	Exchange a valid refresh token for a new access token. Rate-limited (30 / 15 min).
4	POST	/forgot-password	 Public	Send a password-reset OTP to the user's registered email.
5	POST	/reset-password	 Public	Verify OTP and set new password.
6	POST	/change-password	 Any Authenticated	Change password while logged in (requires current password).

3. Users Module

Why: Manage internal staff accounts (ADMIN, GM, TM, TO, SURVEYOR) and client user accounts.

Base Path: /api/v1/users

#	Method	Endpoint	Access	Description
1	GET	/me	 Any Authenticated	Get own profile details.
2	PUT	/me	 Any Authenticated	Update own profile (name, phone, etc.).
3	PUT	/profile-pic	 Any Authenticated	Upload a new profile picture (multipart).
4	PUT	/fcm-token	 Any Authenticated	Register/update Firebase Cloud Messaging token for push notifications.

#	Method	Endpoint	Access	Description
5	GET	/	ADMIN	List all users in the system.
6	POST	/	ADMIN	Create a new user account (staff or client).
7	PUT	/:id	ADMIN	Update another user's details (name, role, email).
8	PUT	/:id/status	ADMIN	Activate, suspend, or deactivate a user account.
9	DELETE	/:id	ADMIN	Permanently delete a user record.

4. Clients Module

Why: Manage client company profiles. A "Client" is a shipping company that owns vessels and requests inspections.

Base Path: `/api/v1/clients`

#	Method	Endpoint	Access	Description
1	GET	/profile/documents	CLIENT	Client views their own company documents.
2	POST	/	ADMIN, GM, TM	Register a new client company.
3	GET	/	ADMIN, GM, TM, TO	List all client companies.
4	GET	/:id	ADMIN, GM, TM, TO	Get details of a specific client company.
5	GET	/:id/documents	ADMIN, GM, TM, TO	View documents uploaded by a specific client.
6	PUT	/:id	ADMIN, GM, TM	Update client company details.
7	DELETE	/:id	ADMIN	Delete a client company record.

5. Vessels Module

Why: Manage the fleet of vessels that are inspected and certified. Every job is tied to a vessel.

Base Path: `/api/v1/vessels`

#	Method	Endpoint	Access	Description
1	GET	/	ADMIN, GM, TM, TO, CLIENT	List vessels. Clients see only their own fleet.
2	GET	/client/:clientId	ADMIN, GM, TM	Get all vessels owned by a specific client company.
3	POST	/	ADMIN, GM, TM	Register a new vessel (IMO number, name, type, etc.).
4	GET	/:id	ADMIN, GM, TM, TO, SURVEYOR, CLIENT	Get full details of a specific vessel.
5	PUT	/:id	ADMIN, GM, TM	Update vessel specifications.

6. Jobs Module

Why: The core workflow engine. A "Job" represents a request for vessel inspection/certification. It flows through statuses: `CREATED` → `DOCUMENT_VERIFIED` → `APPROVED` → `ASSIGNED` → `SURVEY_AUTHORIZED` → `IN_PROGRESS` → `SURVEY_DONE` → `REVIEWED` → `FINALIZED` → `PAYMENT_DONE` → `CERTIFIED`.

Base Path: `/api/v1/jobs`

#	Method	Endpoint	Access	Description
1	GET	/	CLIENT, ADMIN, GM, TM,	List jobs. Clients see own vessels only; Surveyors see assigned jobs only. Supports filters: <code>status</code> ,

#	Method	Endpoint	Access	Description
			TO, SURVEYOR	vessel_id , certificate_type_id , created_from , created_to .
2	GET	/:id	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	Full job details including vessel info, status history, survey data, payment status, and certificate link.
3	POST	/	CLIENT, ADMIN, GM	Create a new inspection request. Validates mandatory documents and vessel ACTIVE status.
4	GET	/:id/eligible-surveyors	ADMIN, GM, TM	Returns surveyors matching the job's vessel type and certificate authority requirements. Shows eligible vs ineligible with reasons.
5	PUT	/:id/verify-documents	TO	Technical Officer verifies uploaded documents. Moves: CREATED → DOCUMENT_VERIFIED .
6	PUT	/:id/approve-request	GM	Business approval. Records approved_by_user_id . Moves: DOCUMENT_VERIFIED → APPROVED .
7	PUT	/:id/finalize	GM, TM	Closes job for certification. Blocks if any NC is still OPEN. Moves → FINALIZED .
8	PUT	/:id/assign	GM	Assigns a surveyor. Runs Authorization Matrix check

#	Method	Endpoint	Access	Description
				(vessel type + cert authority + availability). Moves: APPROVED → ASSIGNED .
9	PUT	<code>/:id/reassign</code>	GM	Swaps surveyor without changing job status. Syncs the linked survey record too.
10	PUT	<code>/:id/reschedule</code>	GM	Changes target port/date. Logs the old values. Notifies assigned surveyor. Only allowed before IN_PROGRESS .
11	PUT	<code>/:id/authorize-survey</code>	TM, GM, ADMIN	Final operational clearance for surveyor. Moves: ASSIGNED → SURVEY_AUTHORIZED .
12	PUT	<code>/:id/review</code>	TO	Post-survey technical review. Moves: SURVEY_DONE → REVIEWED .
13	PUT	<code>/:id/send-back</code>	TM, TO	Request surveyor to redo their work. Moves → REWORK_REQUESTED .
14	PUT	<code>/:id/reject</code>	GM, TM	Terminates the job. GM can reject at CREATED only; TM at ASSIGNED , SURVEY_DONE , REVIEWED .
15	PUT	<code>/:id/cancel</code>	CLIENT, GM, TM, ADMIN	Withdraw/cancel a job request. Cannot cancel terminal-state jobs.
16	PUT	<code>/:id/priority</code>	GM, TM	Set job priority (URGENT, HIGH, NORMAL, LOW).

#	Method	Endpoint	Access	Description
17	GET	<code>/:id/history</code>	ADMIN, GM, TM, TO	Full audit trail of every status change (who, when, reason).
18	POST	<code>/:id/notes</code>	ADMIN, GM, TM, TO	Add private internal notes (not visible to Client).
19	GET	<code>/:id/messages/external</code>	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	Shared chat visible to all parties including Client.
20	GET	<code>/:id/messages/internal</code>	ADMIN, GM, TM, TO	Private staff-only messaging. Client CANNOT see these.
21	POST	<code>/:id/messages</code>	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	Send a message with optional file attachment.

7. Surveys Module

Why: Manages the physical inspection execution done by surveyors on-site. Tracks GPS, checklists, evidence, and final report submission.

Base Path: `/api/v1/surveys`

#	Method	Endpoint	Access	Description
1	POST	<code>/start</code>	SURVEYOR	Check-in at the vessel. Records start GPS and timestamp. Job must be SURVEY_AUTHORIZED.
2	POST	<code>/jobs/:jobId/proof</code>	SURVEYOR	Upload evidence proof (photo/document).

#	Method	Endpoint	Access	Description
				Survey must have checklist submitted first.
3	POST	/jobs/:jobId/location	SURVEYOR	Stream live GPS coordinates during the inspection.
4	GET	/jobs/:jobId/signed-checklist-upload-url	SURVEYOR	Get pre-signed S3 URL for uploading a signed/stamped checklist scan.
5	PUT	/jobs/:jobId/signed-checklist	SURVEYOR	Save the S3 keys of the uploaded signed checklist documents.
6	POST	/jobs/:jobId/sync	SURVEYOR	Offline sync — batch upload checklist answers and GPS points captured without connectivity.
7	POST	/	SURVEYOR	Final survey report submission. Requires GPS, attendance photo, and generates an immutable SHA-256 declaration hash.
8	PUT	/jobs/:jobId/finalize	TM	Finalize survey. Checks all NCs are closed. Separation-of-duties enforced.
9	PUT	/jobs/:jobId/rework	GM, TM	Request the surveyor to redo the survey. Moves survey → REWORK_REQUIRED .
10	POST	/jobs/:jobId/violation	TM	Flag suspicious behavior. Logs to audit trail and alerts ADMIN.

#	Method	Endpoint	Access	Description
11	POST	/jobs/:jobId/statement/draft	SURVEYOR, TM	Draft a survey statement (findings text). TM can trigger background PDF generation.
12	POST	/jobs/:jobId/statement/issue	TM	Issue the official survey statement with signed PDF.
13	GET	/	ADMIN, GM, TM, TO	List all survey reports with pagination and filters.
14	GET	/jobs/:jobId/timeline	ADMIN, GM, TM, TO, SURVEYOR	Full execution timeline with GPS trace and status history.
15	GET	/jobs/:jobId	ADMIN, GM, TM, TO, SURVEYOR	Get survey details for a specific job.

8. Checklists Module

Why: Manages the digital inspection checklist that surveyors fill during a survey.

Base Path: /api/v1/checklists

#	Method	Endpoint	Access	Description
1	GET	/jobs/:jobId	 Any Authenticated	Get the checklist questions and answers for a specific job.
2	PUT	/jobs/:jobId	SURVEYOR	Submit/update checklist answers.
3	GET	/jobs/:jobId/get-upload-url	SURVEYOR	Get pre-signed S3 URL for uploading checklist-related files.

9. Checklist Templates Module

Why: Admin-managed templates that define what checklist questions a surveyor must fill for each certificate type.

Base Path: `/api/v1/checklist-templates`

#	Method	Endpoint	Access	Description
1	POST	/	ADMIN	Create a new checklist template with questions.
2	GET	/get-upload-url	ADMIN	Get pre-signed URL for uploading template PDFs.
3	GET	/	ADMIN, GM, TM, SURVEYOR	List all templates. Filterable by <code>status</code> and <code>certificate_type_id</code> .
4	GET	/job/:jobId/download	SURVEYOR, ADMIN, GM, TM, TO	Download auto-filled checklist DOCX for a specific job.
5	GET	/job/:jobId	SURVEYOR, ADMIN, GM, TM, TO	Get the template assigned to a specific job's certificate type.
6	GET	/:id	ADMIN, GM, TM, SURVEYOR	Get a specific template by ID.
7	PUT	/:id	ADMIN	Update template details and questions.
8	PUT	/:id/activate	ADMIN	Activate or deactivate a template.
9	POST	/:id/clone	ADMIN	Clone an existing template to create a new version.
10	DELETE	/:id	ADMIN	Soft-delete a checklist template.

10. Certificates Module

Why: Manages the full lifecycle of maritime certificates — from draft generation to issuance, renewal, suspension, and revocation.

Base Path: `/api/v1/certificates`

#	Method	Endpoint	Access	Description
1	GET	<code>/verify/:number</code>	 Public	Verify a certificate by its number. No authentication required.
2	GET	<code>/types</code>	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	List all certificate types (e.g., Safety, Cargo, Load Line).
3	POST	<code>/types</code>	ADMIN	Create a new certificate type definition.
4	GET	<code>/types/:id</code>	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	Get details of a specific certificate type.
5	PUT	<code>/types/:id</code>	ADMIN, TM	Update certificate type metadata.
6	GET	<code>/types/:id/required-documents</code>	ADMIN, TM	List mandatory documents for a certificate type.
7	POST	<code>/types/:id/required-documents</code>	ADMIN, TM	Add a mandatory document requirement.
8	PUT	<code>/types/:id/required-documents/:docId</code>	ADMIN, TM	Update a mandatory document requirement.

#	Method	Endpoint	Access	Description
9	DELETE	/types/:id/required-documents/:docId	ADMIN, TM	Remove a mandatory document requirement.
10	GET	/	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	List all certificates. Clients see only their own.
11	GET	/expiring	CLIENT, ADMIN, GM, TM, TO	Get certificates expiring within a date range.
12	GET	/upload-url	ADMIN, GM, TM	Get pre-signed S3 URL for uploading certificate-related files.
13	POST	/vessel/:vesselId/external	ADMIN, GM, TM	Upload an externally-issued certificate for a vessel.
14	GET	/vessel/:vesselId	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	Get all certificates for a specific vessel.
15	GET	/job/:jobId	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	Get the certificate generated from a specific job.
16	POST	/	TM, GM	Generate a new certificate DRAFT for a finalized job.
17	PUT	/:id	TM, GM	Update draft certificate details before issuance.

#	Method	Endpoint	Access	Description
18	POST	<code>/:id/issue</code>	GM	ISSUE the certificate. Generates the final PDF with QR code.
19	GET	<code>/:id</code>	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	Get specific certificate details.
20	GET	<code>/:id/download</code>	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	Download the certificate PDF file.
21	PUT	<code>/:id/suspend</code>	TM	Temporarily suspend a certificate.
22	PUT	<code>/:id/revoke</code>	TM	Permanently revoke a certificate.
23	PUT	<code>/:id/restore</code>	TM	Restore a suspended certificate back to active.
24	PUT	<code>/:id/renew</code>	TM	Renew an expiring certificate with new dates.
25	POST	<code>/bulk-renew</code>	TM	Batch renew multiple certificates at once.
26	POST	<code>/:id/reissue</code>	TM	Reissue a certificate (increments

#	Method	Endpoint	Access	Description
				version, revokes old).
27	GET	<code>/:id/preview</code>	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	Preview the certificate without downloading.
28	GET	<code>/:id/history</code>	CLIENT, ADMIN, GM, TM, TO, SURVEYOR	View full version history and status change audit trail.
29	POST	<code>/:id/transfer</code>	GM	Transfer a certificate to a different vessel (ownership change).
30	POST	<code>/:id/extend</code>	GM	Extend certificate validity period.
31	PUT	<code>/:id/downgrade</code>	GM	Downgrade certificate conditions.

11. Certificate Authorities Module

Why: Manage the issuing authorities (organizations) that are referenced on certificates (e.g., Flag States, IMO bodies).

Base Path: `/api/v1/certificates/authorities`

#	Method	Endpoint	Access	Description
1	GET	<code>/</code>	ADMIN, GM	List all certificate authorities.
2	POST	<code>/</code>	ADMIN	Register a new certificate authority.

#	Method	Endpoint	Access	Description
3	GET	/upload-logo	ADMIN	Get pre-signed S3 URL for authority logo upload.
4	GET	/:id	ADMIN, GM	Get details of a specific authority.
5	PUT	/:id	ADMIN	Update authority details.
6	DELETE	/:id	ADMIN	Delete a certificate authority record.

12. Non-Conformities (NC) Module

Why: Track regulatory failures found during inspections. An open NC blocks job finalization and certificate issuance.

Base Path: /api/v1/non-conformities

#	Method	Endpoint	Access	Description
1	POST	/	SURVEYOR, TO	Raise a new NC during or after a survey. Triggers alerts to TM and TO.
2	GET	/	ADMIN, GM, TM, TO	List all NCs with filters (job_id , status).
3	GET	/:id	ADMIN, GM, TM, TO, SURVEYOR	Get details of a specific NC.
4	PUT	/:id/close	TO, TM	Close an NC after corrective action is verified. Records closure_remarks and closed_at .
5	GET	/job/:jobId	ADMIN, GM, TM, TO, SURVEYOR	Get all NCs linked to a specific job.

13. Payments Module

Why: Financial tracking for invoicing, payments, and accounting related to inspection services.

Base Path: `/api/v1/payments`

#	Method	Endpoint	Access	Description
1	GET	/	CLIENT, ADMIN, GM, TM	List all payments/invoices. Clients see only their own.
2	GET	/summary	CLIENT, ADMIN, GM	Financial summary (total paid, outstanding, overdue).
3	GET	/:id	CLIENT, ADMIN, GM, TM	Get specific payment/invoice details.
4	POST	/invoice	ADMIN, GM, TM	Create a new invoice for a job.
5	PUT	/:id/pay	ADMIN, GM, TM	Mark an invoice as paid. Supports receipt file upload.
6	POST	/:id/refund	ADMIN, GM	Process a refund for a paid invoice.
7	POST	/:id/partial	ADMIN, GM, TM	Record a partial payment against an invoice.
8	GET	/:id/ledger	ADMIN, GM	View the financial ledger for a specific payment.
9	POST	/writeoff	ADMIN	Write off an unpaid invoice (accounting).

14. Incidents Module

Why: Report and manage unexpected vessel events, accidents, or technical failures.

Base Path: `/api/v1/incidents`

#	Method	Endpoint	Access	Description
1	POST	/	CLIENT, ADMIN, GM, TM	Report a new incident for a vessel.
2	GET	/	CLIENT, ADMIN, GM, TM, TO	List all incidents. Clients see only their vessels.

#	Method	Endpoint	Access	Description
3	GET	<code>/:id</code>	CLIENT, ADMIN, GM, TM, TO	Get details of a specific incident.
4	PUT	<code>/:id/status</code>	ADMIN, GM, TM	Update incident status and add remarks.

15. Activity Requests Module

Why: Handle operational activity requests like additional inspections, schedule changes, or special surveys.

Base Path: `/api/v1/activity-requests`

#	Method	Endpoint	Access	Description
1	POST	<code>/</code>	CLIENT, ADMIN, GM, TM	Create a new activity request.
2	GET	<code>/</code>	CLIENT, ADMIN, GM, TM, TO	List all requests.
3	GET	<code>/:id</code>	CLIENT, ADMIN, GM, TM, TO	Get details of a specific request.
4	PUT	<code>/:id/status</code>	ADMIN, GM, TM	Approve, reject, or update request status.

16. Change Requests Module

Why: Formal requests to modify existing job data, certificates, or vessel records through an approval flow.

Base Path: `/api/v1/change-requests`

#	Method	Endpoint	Access	Description
1	POST	<code>/</code>	CLIENT, ADMIN, GM, TM	Submit a change request.

#	Method	Endpoint	Access	Description
2	GET	/	ADMIN, GM, TM	List all pending change requests.
3	GET	/:id	ADMIN, GM, TM, CLIENT	Get change request details.
4	PUT	/:id/approve	ADMIN, GM	Approve the change request.
5	PUT	/:id/reject	ADMIN, GM	Reject the change request.

17. Approvals Module

Why: Generic multi-step workflow approval engine used across various modules.

Base Path: /api/v1/approvals

#	Method	Endpoint	Access	Description
1	POST	/	ADMIN, GM, TM	Create a new approval workflow record.
2	PUT	/:id/step	ADMIN, GM, TM	Advance or update a specific step in the approval chain.

18. TOCA Module

Why: Manages Transfer of Class Authority — when a vessel's classification responsibility transfers between organizations.

Base Path: /api/v1/toca

#	Method	Endpoint	Access	Description
1	POST	/	TM	Create a new TOCA record.
2	PUT	/:id/status	TM, ADMIN	Update TOCA status (approve, reject, finalize).
3	GET	/	ADMIN, GM, TM	List all TOCA records.

19. Surveyors Module

Why: Manage surveyor profiles, applications, authorizations, availability, and GPS tracking.

Base Path: `/api/v1/surveyors`

#	Method	Endpoint	Access	Description
1	GET	<code>/get-upload-url</code>	 Public	Get pre-signed URLs for uploading application documents. Rate-limited.
2	POST	<code>/apply</code>	 Public	Submit a surveyor application with credentials and documents. Rate-limited.
3	GET	<code>/</code>	ADMIN, GM, TM	List all registered surveyors.
4	POST	<code>/</code>	ADMIN, TM	Manually create a new surveyor profile.
5	GET	<code>/applications</code>	ADMIN, TM	List pending surveyor applications for review.
6	PUT	<code>/applications/:id/review</code>	TM, ADMIN	Approve or reject a surveyor application.
7	PUT	<code>/:id/status</code>	ADMIN, TM	Suspend or activate a surveyor.
8	GET	<code>/:id/profile</code>	ADMIN, TM, SURVEYOR, GM	View surveyor profile (certifications, authorized ship types, etc.).
9	PUT	<code>/:id/profile</code>	ADMIN, TM	Update surveyor profile and

#	Method	Endpoint	Access	Description
				authorizations.
10	POST	/availability	SURVEYOR	Toggle availability status (online/offline).
11	POST	/location	SURVEYOR	Report current GPS location.
12	GET	/:id/location-history	ADMIN, TM, GM	View GPS history for a specific surveyor.

20. Documents Module

Why: Centralized document management system — upload, view, and manage files attached to any entity (vessel, job, client, etc.).

Base Path: /api/v1/documents

#	Method	Endpoint	Access	Description
1	GET	/get-upload-url	CLIENT, ADMIN, GM, TM, SURVEYOR	Get a pre-signed S3 URL for direct browser upload.
2	POST	/upload	CLIENT, ADMIN, GM, TM, SURVEYOR	Upload a standalone file via multipart.
3	POST	/register	CLIENT, ADMIN, GM, TM, SURVEYOR	Register an already-uploaded S3 key as a document.
4	GET	/:id	All Roles	Get document metadata and download URL.

#	Method	Endpoint	Access	Description
5	GET	<code>/:entityType/:entityId</code>	All Roles	List all documents for an entity (e.g., <code>/vessel/123</code> or <code>/job/456</code>).
6	POST	<code>/:entityType/:entityId</code>	CLIENT, ADMIN, GM, TM, SURVEYOR	Upload documents directly attached to an entity.
7	POST	<code>/:entityType/:entityId/register</code>	CLIENT, ADMIN, GM, TM, SURVEYOR	Register pre-uploaded files to an entity.
8	DELETE	<code>/:id</code>	ADMIN, GM	Delete a document record.

21. Feedback Module

Why: Captures client satisfaction feedback after a job completion.

Base Path: `/api/v1/feedback`

#	Method	Endpoint	Access	Description
1	POST	<code>/</code>	CLIENT	Submit feedback/rating for a completed job.
2	GET	<code>/</code>	ADMIN, GM	List all submitted feedback entries.
3	GET	<code>/:id</code>	ADMIN, GM	Get a specific feedback entry.
4	GET	<code>/job/:jobId</code>	ADMIN, GM, CLIENT	Get feedback for a specific job.

22. Portfolio Feedback Module

Why: Manages public-facing testimonials/reviews displayable on the company portfolio website.

Base Path: `/api/v1/portfolio-feedback`

#	Method	Endpoint	Access	Description
1	GET	<code>/public</code>	 Public	Get all visible testimonials for the portfolio website.
2	POST	<code>/</code>	CLIENT	Submit or update portfolio feedback/testimonial.
3	GET	<code>/my-feedback</code>	CLIENT	Client views their own submitted testimonial.
4	GET	<code>/</code>	ADMIN, GM	List all portfolio feedback entries (visible + hidden).
5	PATCH	<code>/:id/visibility</code>	ADMIN	Toggle whether a testimonial appears on the public website.

23. Notifications Module

Why: Push notification management — lists, reads, and clears notifications for the logged-in user.

Base Path: `/api/v1/notifications`

#	Method	Endpoint	Access	Description
1	GET	<code>/</code>	 Any Authenticated	Get own notifications (paginated).
2	PUT	<code>/:id/read</code>	 Any Authenticated	Mark a specific notification as read.
3	PUT	<code>/read-all</code>	 Any Authenticated	Mark all notifications as read.

24. Dashboard Module

Why: Returns role-specific aggregated statistics and KPIs for the home dashboard.

Base Path: `/api/v1/dashboard`

#	Method	Endpoint	Access	Description
1	GET	/	All Roles	Returns dashboard data customized per role (e.g., Client sees their vessel count, GM sees pending approvals, Surveyor sees assigned jobs).

25. Search Module

Why: Global cross-module search — finds jobs, vessels, certificates, and clients from a single query.

Base Path: `/api/v1/search`

#	Method	Endpoint	Access	Description
1	GET	/	 Any Authenticated	Global search across all entities. Results are scoped by user role.

26. Reports Module

Why: Generate analytical reports for management oversight — certificate stats, surveyor performance, NC trends, and financials.

Base Path: `/api/v1/reports`

#	Method	Endpoint	Access	Description
1	GET	/certificates	ADMIN, GM, TM	Certificate issuance, expiry, and status distribution report.
2	GET	/surveyors	ADMIN, GM, TM	Surveyor performance and workload report.
3	GET	/non-conformities	ADMIN, GM, TM	NC trends, severity distribution, and resolution time report.

#	Method	Endpoint	Access	Description
4	GET	/financials	ADMIN, GM, TM	Revenue, invoicing, and payment collection report.

27. Flags Module

Why: Manage maritime flag administrations (country registries that vessels are registered under).

Base Path: /api/v1/flags

#	Method	Endpoint	Access	Description
1	POST	/	ADMIN	Create a new flag administration entry.
2	GET	/	ADMIN, GM, TM, TO	List all flag administrations.
3	GET	/:id	ADMIN, GM, TM, TO	Get details of a specific flag.
4	PUT	/:id	ADMIN	Update a flag administration entry.

28. Templates Module

Why: Manage reusable checklist/form templates for inspections.

Base Path: /api/v1/templates

#	Method	Endpoint	Access	Description
1	POST	/	ADMIN	Create a new template.
2	GET	/	ADMIN, GM, TM	List all templates.
3	GET	/:id	ADMIN, GM, TM	Get template details.
4	PUT	/:id	ADMIN	Update template.
5	DELETE	/:id	ADMIN	Delete a template.

29. Support Module

Why: Internal ticketing system for users to report issues, request help, or submit queries.

Base Path: `/api/v1/support`

#	Method	Endpoint	Access	Description
1	POST	/	 Any Authenticated	Create a new support ticket.
2	GET	/	 Any Authenticated	List own support tickets.
3	GET	/:id	 Any Authenticated	Get details of a specific ticket.
4	PUT	/:id/status	ADMIN, GM	Update ticket status (resolve, close, escalate).
5	PUT	/:id	ADMIN, GM	Update ticket details.

30. Compliance Module

Why: GDPR and data governance — export user data or anonymize records upon request.

Base Path: `/api/v1/compliance`

#	Method	Endpoint	Access	Description
1	GET	/export/:id	ADMIN, CLIENT	Export all data related to a user (GDPR data subject request).
2	POST	/anonymize/:id	ADMIN	Anonymize a user's personal data (right to erasure).

31. System Module

Why: Health checks, audit logs, and system administration tools for DevOps and ADMIN users.

Base Path: `/api/v1/system`

#	Method	Endpoint	Access	Description
1	GET	/health	 Any Authenticated	System health check endpoint.
2	GET	/readiness	 Any Authenticated	Readiness probe (DB, Redis, S3 connectivity).
3	GET	/version	 Any Authenticated	Current API version and build info.
4	GET	/metrics	ADMIN	System performance metrics.
5	GET	/audit-logs	ADMIN	View all audit log entries.
6	POST	/users/:id/logout	ADMIN	Force logout a specific user (invalidate their tokens).
7	GET	/migrations	ADMIN	View database migration status.
8	GET	/jobs/failed	ADMIN	View failed background jobs.
9	POST	/jobs/:id/retry	ADMIN	Retry a failed background job.
10	POST	/maintenance/:action	ADMIN	Trigger maintenance actions (cache clear, etc.).
11	GET	/feature-flags	ADMIN	View feature flag configuration.
12	GET	/locales	ADMIN	View available localization strings.

32. Public Module

Why: Unauthenticated endpoints for external verification and website content.

Base Path: /api/v1/public

#	Method	Endpoint	Access	Description
1	GET	/certificate/verify/:number	 Public	Verify a certificate's authenticity by number.
2	GET	/vessel/:imo	 Public	Look up a vessel by IMO number.
3	GET	/website/videos	 Public	Get portfolio showcase videos.

33. Website / CMS Module

Why: Content management for the public-facing portfolio website — videos and static page management.

Base Path: /api/v1/website

#	Method	Endpoint	Access	Description
1	GET	/videos	 Public	List published portfolio videos.
2	POST	/videos	ADMIN	Upload a new showcase video with thumbnail.
3	PUT	/videos/:id	ADMIN	Update video details or replace file.
4	DELETE	/videos/:id	ADMIN	Delete a video.

34. Newsletter Module

Why: Email newsletter subscription management and broadcast for the portfolio website.

Base Path: /api/v1/website/newsletter

#	Method	Endpoint	Access	Description
1	POST	/subscribe	 Public	Subscribe to newsletter with email.

#	Method	Endpoint	Access	Description
2	POST	/unsubscribe	 Public	Unsubscribe from newsletter.
3	GET	/unsubscribe-one-click	 Public	RFC 8058 one-click unsubscribe (GET).
4	POST	/unsubscribe-one-click	 Public	RFC 8058 one-click unsubscribe (POST).
5	GET	/subscribers	ADMIN	List all newsletter subscribers.
6	POST	/send	ADMIN	Broadcast a newsletter to all subscribers.

35. Site Static Content Module

Why: CMS for static pages on the portfolio website — FAQ, About Us, Terms of Service, Privacy Policy, etc.

Base Path: /api/v1/website/static-content

#	Method	Endpoint	Access	Description
1	GET	/	 Public (optional auth)	List published static pages. ADMINS also see draft pages.
2	GET	/:slug	 Public (optional auth)	Get a page by URL slug. ADMINS see unpublished pages too.
3	POST	/	ADMIN	Create a new static page.
4	PUT	/:slug	ADMIN	Update a page by slug.
5	DELETE	/:slug	ADMIN	Delete a page by slug.
6	GET	/admin/:id	ADMIN	Get a page by ID (admin view).
7	PUT	/admin/:id	ADMIN	Update a page by ID (admin view).
8	DELETE	/admin/:id	ADMIN	Delete a page by ID (admin view).

36. Contact Module

Why: "Contact Us" form from the portfolio website — enquiry submission and admin management.

Base Path: `/api/v1/contact`

#	Method	Endpoint	Access	Description
1	POST	/	 Public	Submit a contact enquiry (name, email, message). Sends auto-acknowledgement email.
2	GET	/stats	ADMIN, GM	Get enquiry statistics (total, new, responded).
3	GET	/	ADMIN, GM	List all enquiries with pagination.
4	GET	/:id	ADMIN, GM	Get enquiry details.
5	PATCH	/:id/status	ADMIN, GM	Update enquiry status (new → responded → closed).
6	DELETE	/:id	ADMIN	Delete an enquiry.

Legend

Icon	Meaning
 Public	No authentication required
 Any Authenticated	Any logged-in user
ROLE_NAME	Only these specific roles can access